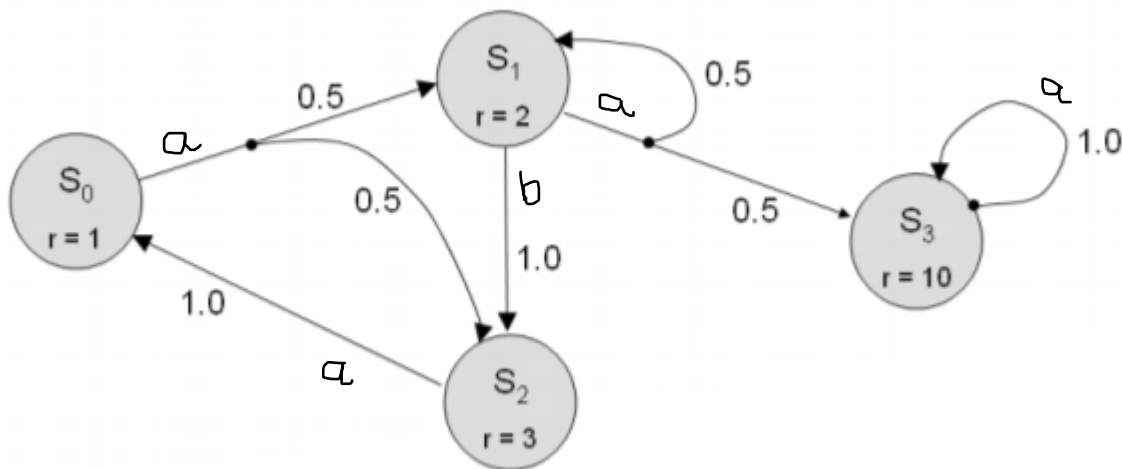


Markov Decision Process

Practice Problems

QUESTION 1

Consider the MDP given in the figure below. Assume the discount factor $\gamma = 0.9$. The r -values (written within the node) are rewards that the agent gets when in that node. The numbers next to arrows are probabilities of outcomes. Note that only state S_1 has two actions (a and b). The other states have only one action (a) for each state.



Write down the optimal value of state S_1 . (Hint: first find the optimal policy for S_3 using the closed form solution. Assume the optimal policy from state S_1 is to take action a)

Sol: $V(S_3) = 10 + 0.9 \cdot V(S_3)$

$$V(S_3) = 100$$

$$V(S_1) = 2 + \max (0.9 (0.5 \cdot V(S_1) + 0.5 \cdot V(S_3)), 0.9 (V(S_2)))$$

$$V(S_1) = 2 + 0.9 (0.5 \cdot V(S_1) + 50)$$

$$V(S_1) = 2 + 0.45 \cdot V(S_1) + 45$$

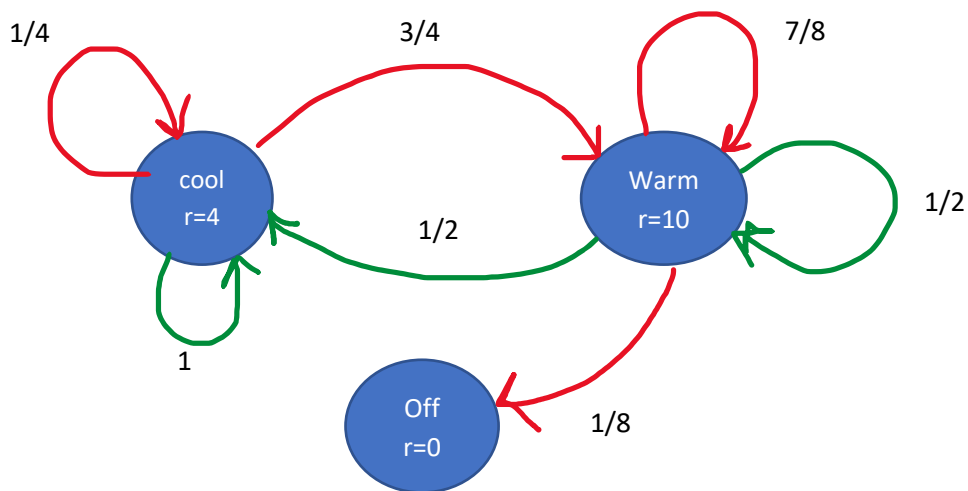
$$0.55 \cdot V(S_1) = 47$$

$$V(S_1) = 85.45$$

Since the optimal action from state s_1 is 'a' therefore $\text{term1} > \text{term2}$ in $\max$

QUESTION 2

Consider a robot that is moving in an environment. The goal of the robot is to move from an initial point to a destination point as fast as possible. However, the robot has the limitation that if it moves fast, its engine can overheat and stop the robot from moving. The robot can move with two different speeds: slow and fast. We can model this problem as an MDP by having three states: **cool**, **warm**, and **off**. The transitions are shown below (fast: red, slow: green). The reward of being in cool state is 4 and the reward of being in warm state is 10. Assume that the discount factor is 0.9 and, also assume that when we reach the state off, we remain there without getting any reward.



s	a	s'	$T(s,a,s')$
cool	slow	cool	1
cool	fast	cool	1/4
cool	fast	warm	3/4
warm	slow	cool	1/2
warm	slow	warm	1/2
warm	fast	warm	7/8
warm	fast	off	1/8

- a. Consider the conservative policy when the robot always moves slowly. What is the value of $V(\text{cool})$ under the conservative policy? Remember that $V(s)$ is the expected discounted sum of rewards when starting at state s

$$V_{\text{slow}}(\text{cool}) = 4 + 0.9 * (V_{\text{slow}}(\text{cool}))$$

$$V_{\text{slow}}(\text{cool}) - 0.9 * (V_{\text{slow}}(\text{cool})) = 4$$

$$V_{\text{slow}}(\text{cool}) = 40$$

- b. What is the optimal policy after two iterations of value iteration. Initialize $V(s)=0$ for all states?

$$V(\text{cool}) = V(\text{warm}) = V(\text{off}) = 0$$

Iter 1:

$$V(\text{cool}) = 4 + 0.9 * \max\left(V(\text{cool}), \frac{1}{4}V(\text{cool}) + \frac{3}{4}V(\text{warm})\right)$$

$$V(\text{cool}) = 4$$

$$V(\text{warm}) = 10 + 0.9 * \max\left(\frac{1}{2}V(\text{cool}) + \frac{1}{2}V(\text{warm}), \frac{7}{8}V(\text{warm}) + \frac{1}{8}V(\text{off})\right)$$

$$V(\text{warm}) = 10$$

$$V(\text{off}) = 0$$

Iter 2:

$$V(\text{cool}) = 4 + 0.9 * \max\left(V(\text{cool}), \frac{1}{4}V(\text{cool}) + \frac{3}{4}V(\text{warm})\right)$$

$$V(\text{cool}) = 4 + 0.9 * \max\left(4, \frac{1}{4}4 + \frac{3}{4}10\right)$$

$$V(\text{cool}) = 11.65$$

Use values of $V(s)$ from previous state

$$V(\text{warm}) = 10 + 0.9 * \max\left(\frac{1}{2}4 + \frac{1}{2}10, \frac{7}{8}10 + \frac{1}{8}0\right)$$

$$V(\text{warm}) = 17.875$$

The best policy after iter 2:

$$\pi^* = \underset{a \in A(s)}{\operatorname{argmax}} \sum_{s'} T(s, a, s') V(s')$$

$$\pi^*(cool) = \underset{a \in A(s)}{\operatorname{argmax}} \left[V(cool), \frac{1}{4} V(cool) + \frac{3}{4} V(warm) \right] = \underset{a \in A(s)}{\operatorname{argmax}} \left[11.65, \frac{1}{4} 11.65 + \frac{3}{4} 17.875 \right] = \underset{a \in A(s)}{\operatorname{argmax}} [11.65, 16.31]$$

$$\pi^*(cool) = fast$$

$$\pi^*(warm) = \underset{a \in A(s)}{\operatorname{argmax}} \left[\frac{1}{2} V(cool) + \frac{1}{2} V(warm), \frac{7}{8} V(warm) \right] = \underset{a \in A(s)}{\operatorname{argmax}} [14.76, 15.64]$$

$$\pi^*(warm) = fast$$

NOTE: value iteration has not converged, so the optimal policy after iteration 2 is not guaranteed to be optimal. After the algorithm has converged, the optimal value and policy is as follows:

{'warm': 77.95918362113281, 'cool': 73.06122443745934, 'off': 0.0}

{'warm': 'slow', 'cool': 'fast', 'off': None}

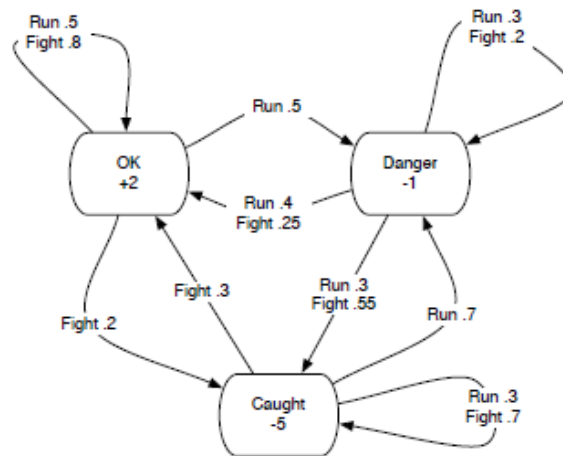
QUESTION 3

1. A boy is being chased around the school yard by bullies and must choose whether to Fight or Run.

- There are three states:
 - Ok (O), where he is fine for the moment.
 - Danger (D), where the bullies are right on his heels.
 - Caught (C), where the bullies catch up with him and administer noogies.

use $\gamma = 0.9$

The graph of the MDP is given here:



The reward of taking any action from a state is written within the state

1. Fill out the table after running the value-iteration algorithm for two iterations

k	$V_{opt}(O)$	$V_{opt}(D)$	$V_{opt}(C)$
1	2	-1	-5
2	2.54	-1.9	-6.98

Iter 1:

$V(O)=2$

$V(D)=-1$

$V(C)=-5$

Iter 2:

$$\begin{aligned}
 \underline{V_{opt}(O)} &= \max(2 + .9((.5 * 2) + (.5 * -1)), 2 + .9((.8 * 2) + (.2 * -5))) \\
 &= \max(2.45, 2.54) \\
 &= 2.54
 \end{aligned}$$

$$\begin{aligned}
 \underline{V_{opt}(D)} &= \max(-1 + .9((.4 * 2) + (.3 * -1) + (.3 * -5)), -1 + .9((.25 * 2) + (.2 * -1) + (.55 * -1))) \\
 &= \max(-1.9, -3.2050) \\
 &= -1.9
 \end{aligned}$$

$$\begin{aligned}
 \underline{V_{opt}(C)} &= \max(-5 + .9((.7 * -1) + (.3 * -5)), -5 + .9((.3 * 2) + (.7 * -5))) \\
 &= \max(-6.98, -7.61) \\
 &= -6.98
 \end{aligned}$$

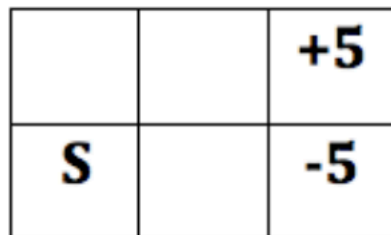
2. At $k = 2$ with $\gamma = .9$ what policy would you select? Is it necessarily true that this is the optimal policy? At $k = 3$ what policy would you select? Is it necessarily true that this is the optimal policy?

- From O choose Fight.
- From D choose Run.
- From C choose Run.

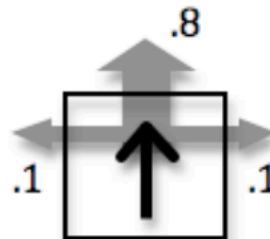
Value iteration that has not converged is not guaranteed to find the optimal policy, so this policy is not necessarily optimal.

QUESTION 4

This gridworld MDP operates like to the one we saw in class. The states are grid squares, identified by their row and column number (row first). The agent always starts in state (1,1), marked with the letter S. There are two terminal goal states, (2,3) with reward +5 and (1,3) with reward -5. Rewards are 0 in non-terminal states. The transition function is such that the intended agent movement (North, South, West, or East) happens with probability .8. With probability .1 each, the agent ends up in one of the states perpendicular to the intended direction. If a collision with a wall happens, the agent stays in the same state



(a)



(b)

Figure 1: (a) Gridworld MDP. (b) Transition function.

1. Draw the optimal policy for this grid

S =	(1,1)	(1,2)	(1,3)	(2,1)	(2,2)	(2,3)
$\pi^*(S) =$	Up	Left	NA	Right	Right	NA

2. Give the first two rounds of value iteration updates for each state, with a discount of 0.9. (Assume V_0 is 0 everywhere and compute V_i for times $i = 1, 2, 3$).

All states initialized to zero $V(s) = 0$

Iter 1

$$\begin{aligned}
 V((1,1)) &= 0 & V((2,1)) &= 0 \\
 V((1,2)) &= 0 & V((2,2)) &= 0 \\
 V((1,3)) &= -5 & V((2,3)) &= 5
 \end{aligned}$$

V

0	0	5
0	0	-5

Iter 2

$$V((1,1)) = 0$$

$$V((1,2)) = 0$$

$$V((1,3)) = -5$$

$$V((2,1)) = 0$$

$$V((2,2)) = 0 + 0.9 (0.8 \times 5) = .6$$

$$V((2,3)) = 5$$

V

0	3.6	5
0	0	-5

Iter 3

$$V((1,1)) = 0 + 0.1 \times (-5)$$

$$V((1,2)) = 0.9 \times (0.8 \times 3.6) = 2.14$$

$$V((1,3)) = -5$$

$$V((2,1)) = 0.9 \times (0.8 \times 3.6) = 2.6$$

$$V((2,2)) = 0.9 \times (0.8 \times 5 + 0.1 \times 3.6) = 3.92$$

$$V((2,3)) = 5$$

2.6	3.92	5
↺ ↻ ↻ ↻	↗ 2.14	-5

Optimal policy and its value after iter = 3